CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 25: Complementary Angles

Every formula and result of the chapter on one sheet

1	Complement	$90^\circ - \theta$
2	Sine	$\sin(90^\circ - \theta) = \cos\theta$
3	Cosine	$\cos(90^\circ - \theta) = \sin\theta$
4	Tangent	$\tan(90^\circ - \theta) = \cot\theta$
5	Cotangent	$\cot(90^\circ - \theta) = \tan\theta$
6	Secant	$\sec(90^\circ - \theta) = \operatorname{cosec}\theta$
7	Cosecant	$\operatorname{cosec}(90^\circ - \theta) = \sec\theta$
8	If $A + B = 90^\circ$	$\sin A = \cos B$, $\tan A = \cot B$, $\sec A = \operatorname{cosec} B$
9	Product	$\tan\theta \tan(90^\circ - \theta) = 1$
10	Sum of squares	$\sin^2\theta + \sin^2(90^\circ - \theta) = 1$
11	Triangle	$\frac{B + C}{2} = 90^\circ - \frac{A}{2}$
12	Equation	$\sin A = \cos B$ gives $A + B = 90^\circ$ (acute angles)